

Assignment 7 Solutions

Modern Cryptography (2)

April 17, 2026

Solution 1 (Basic syndrome decoding problem). Let the columns of

$$H = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

be denoted by

$$c_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad c_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad c_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad c_4 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad c_5 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad c_6 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

The target syndrome is

$$s = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

1. We seek all vectors $e \in \mathbb{F}_2^6$ such that

$$He^\top = s, \quad \text{wt}(e) \leq 2.$$

Since $He^\top$ is the sum of the columns selected by the support of e , we look for:

- a single column equal to s , or
- a sum of two columns equal to s .

First, note that

$$c_1 = s,$$

so we obtain one solution of weight 1:

$$e = (1, 0, 0, 0, 0, 0).$$

Next, observe that

$$c_2 + c_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = s,$$

so another solution is

$$e = (0, 1, 1, 0, 0, 0).$$

Also,

$$c_4 + c_6 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = s,$$

so we get a third solution:

$$e = (0, 0, 0, 1, 0, 1).$$

Therefore, all solutions with weight at most 2 are

$$\boxed{(1, 0, 0, 0, 0, 0), \quad (0, 1, 1, 0, 0, 0), \quad (0, 0, 0, 1, 0, 1)}.$$

2. Yes, this syndrome corresponds to a unique minimum-weight error vector. Indeed:

- we found one solution of weight 1, namely

$$(1, 0, 0, 0, 0, 0);$$

- all other solutions we found have weight 2;
- there is no solution of weight 0, since the zero vector has syndrome $0 \neq s$.

Hence the minimum possible weight is 1, and there is exactly one such solution. Therefore

$$\boxed{\text{the syndrome has a unique minimum-weight error vector, namely } (1, 0, 0, 0, 0, 0)}.$$

Solution 2 (Computing the Gilbert–Varshamov bound, including the asymptotic estimate).

1. We must determine the largest integer d such that

$$\sum_{i=0}^{d-2} \binom{40}{i} < 2^{20} = 1,048,576.$$

Compute:

$$\begin{aligned} \binom{40}{0} = 1, \quad \binom{40}{1} = 40, \quad \binom{40}{2} = 780, \quad \binom{40}{3} = 9880, \\ \binom{40}{4} = 91390, \quad \binom{40}{5} = 658008. \end{aligned}$$

Hence

$$\sum_{i=0}^5 \binom{40}{i} = 1 + 40 + 780 + 9880 + 91390 + 658008 = 760099,$$

and this satisfies

$$760099 < 1048576.$$

Therefore $d = 7$ is guaranteed.

Next,

$$\binom{40}{6} = 3838380,$$

so

$$\sum_{i=0}^6 \binom{40}{i} = 760099 + 3838380 = 4598479 > 1048576.$$

Thus $d = 8$ is not guaranteed.

Therefore the largest distance guaranteed by the GV criterion is

$$\boxed{d = 7}.$$

2. For rate

$$R = \frac{1}{2},$$

the asymptotic GV relative distance satisfies

$$H_2(\delta_{\text{GV}}) = \frac{1}{2},$$

where

$$H_2(x) = -x \log_2 x - (1-x) \log_2(1-x).$$

Numerically,

$$\boxed{\delta_{\text{GV}} \approx 0.1100.}$$

3. The finite-length conclusion from part (1) is

$$d = 7, \quad \frac{d}{n} = \frac{7}{40} = 0.175.$$

The asymptotic estimate from part (2) is

$$\delta_{\text{GV}} \approx 0.1100.$$

Thus, in this example, the finite-length relative-distance guarantee is

$$0.175,$$

while the asymptotic estimate is approximately

$$0.1100.$$

This does not create any contradiction: part (1) is a finite-blocklength calculation for the specific parameters $n = 40$ and $k = 20$, whereas part (2) is an asymptotic estimate as $n \rightarrow \infty$.

4. For example, in Python one may compute

```
import math
S = sum(math.comb(40, i) for i in range(6))
print(S)
```

to evaluate

$$\sum_{i=0}^5 \binom{40}{i}.$$

Solution 3 (KZG Commitment). Let

$$f(X) = 3 + 2X + X^2.$$

1. By definition, the KZG commitment is

$$C = g^{f(\tau)}.$$

Since

$$f(\tau) = 3 + 2\tau + \tau^2,$$

we have

$$C = g^{3+2\tau+\tau^2}.$$

Using the public parameters (g, g^τ, g^{τ^2}) , this can be computed as

$$\boxed{C = g^3 \cdot (g^\tau)^2 \cdot g^{\tau^2}.}$$

2. We open at the point

$$z = 1.$$

First compute

$$f(1) = 3 + 2 + 1 = 6.$$

The quotient polynomial is

$$q(X) = \frac{f(X) - f(1)}{X - 1}.$$

Since

$$f(X) - f(1) = X^2 + 2X + 3 - 6 = X^2 + 2X - 3,$$

and

$$X^2 + 2X - 3 = (X - 1)(X + 3),$$

it follows that

$$q(X) = X + 3.$$

The opening proof is

$$\pi = g^{q(\tau)}.$$

Because

$$q(\tau) = \tau + 3,$$

we obtain

$$\boxed{\pi = g^{\tau+3} = g^3 \cdot g^\tau.}$$

3. Let h be a generator of the second pairing group G_2 , and suppose the verifier knows h and h^τ . The verifier checks the pairing equation

$$e(C/g^y, h) \stackrel{?}{=} e(\pi, h^{\tau-z}),$$

where $y = f(z)$.

In our case,

$$z = 1, \quad y = f(1) = 6,$$

so the verification equation becomes

$$\boxed{e(C/g^6, h) \stackrel{?}{=} e(\pi, h^{\tau-1}).}$$

Equivalently, one may write

$$e(C/g^y, h) \stackrel{?}{=} e(\pi, h^\tau/h^z).$$

4. Assume verification succeeds. Then

$$e(C/g^{f(z)}, h) = e(\pi, h^{\tau-z}).$$

Substituting

$$C = g^{f(\tau)}, \quad \pi = g^{q(\tau)},$$

we get

$$e(g^{f(\tau)-f(z)}, h) = e(g^{q(\tau)}, h^{\tau-z}).$$

By bilinearity of the pairing, the right-hand side is

$$e(g, h)^{q(\tau)(\tau-z)},$$

while the left-hand side is

$$e(g, h)^{f(\tau)-f(z)}.$$

Therefore,

$$e(g, h)^{f(\tau)-f(z)} = e(g, h)^{q(\tau)(\tau-z)}.$$

Since $e(g, h)$ generates a prime-order group, the exponents must be equal:

$$\boxed{f(\tau) - f(z) = q(\tau)(\tau - z)}.$$

This is precisely the core of KZG correctness, because it comes from the polynomial identity

$$f(X) - f(z) = (X - z)q(X).$$

Evaluating this identity at the secret point $X = \tau$ yields

$$f(\tau) - f(z) = (\tau - z)q(\tau),$$

which is exactly what the pairing check enforces.

Solution 4 (Merkle tree of size 16).

1. To prove that m_{11} belongs to the Merkle tree, the prover must send the sibling hashes along the path from ℓ_{11} to the root.

Concretely:

- the sibling of ℓ_{11} is ℓ_{12} ;
- the sibling of

$$v_6^{(1)} = H(\ell_{11} \parallel \ell_{12})$$

is $v_5^{(1)}$;

- the sibling of $v_3^{(2)}$ is $v_4^{(2)}$;
- the sibling of $v_2^{(3)}$ is $v_1^{(3)}$.

Hence the authentication path is

$$\boxed{(\ell_{12}, v_5^{(1)}, v_4^{(2)}, v_1^{(3)})}.$$

In addition, the prover must provide a constant number of direction bits indicating whether the current node is a left child or a right child at each level.

2. Since $16 = 2^4$, the tree has height 4. Therefore the membership proof contains exactly 4 sibling hashes.

If each hash has length λ bits, then the proof size is

$$\boxed{4\lambda \text{ bits}}$$

plus a constant number of direction bits.

The complexities are as follows:

- **Building the whole tree:** one computes all leaves and all internal nodes, so the total complexity is

$$\boxed{O(n)}.$$

- **Generating a membership proof for one leaf:** one collects one sibling hash per level, so the complexity is

$$\boxed{O(\log n)}.$$

- **Verifying a membership proof for one leaf:** the verifier recomputes one hash per level up to the root, so the complexity is

$$\boxed{O(\log n)}.$$